

Computer Organization and Architecture · 21CSS201T

Unit 1 · Session 4 · SLO 1 — Solved Worksheet

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Unit – 1 : Session – 4 : SLO – 1 | Number system conversions

■ 5 MCQs

1. a — 11001 ($25 = 16+8+1$)
2. b — 4E ($78 = 4 \times 16 + 14$)
3. c — Octal
4. b — 110110 (groups 110 110 = 6 6)
5. c — 3C ($111100 = 0011\ 1100 = 3\ C$)

■ 2 Two-Mark Questions

1. $10110 = 1 \times 16 + 0 \times 8 + 1 \times 4 + 1 \times 2 + 0 \times 1 = 16 + 4 + 2 = 22$ (decimal).
2. 3F hex: $3 = 0011$, $F = 1111 \rightarrow 00111111$ (binary) = 63 decimal.

■ 1 Scenario-Based Long Answer

- Decimal 78 $\rightarrow$ Binary (repeated division by 2): $78/2=39\ r0$, $39/2=19\ r1$, $19/2=9\ r1$, $9/2=4\ r1$, $4/2=2\ r0$, $2/2=1\ r0$, $1/2=0\ r1$. Reading remainders upward: 1001110.
- Decimal 78 $\rightarrow$ Octal (division by 8): $78/8=9\ r6$, $9/8=1\ r1$, $1/8=0\ r1 \rightarrow 116$ (octal).
- Decimal 78 $\rightarrow$ Hexadecimal (division by 16): $78/16=4\ r14(E)$, $4/16=0\ r4 \rightarrow 4E$.
- Exact values: $78_{10} = 1001110_2 = 116_8 = 4E_{16}$.

■ 5 Match the Following

- Decimal to Binary — c) 2ⁿ weight conversion (repeated division by 2)
- Decimal to Hexadecimal — b) Divide by 16
- Binary to Octal — a) Group digits in 3s
- Hexadecimal to Binary — d) Replace each digit with 4-bit binary
- Octal to Binary — e) Replace each digit with 3-bit binary

■ 5 Fill in the Blanks

1. 1001110
2. base 8
3. 31
4. 4
5. 3C

■ 1 Scenario-Based Small Activity

- Binary 1010110 to octal: pad to groups of three from the right $\rightarrow 001\ 010\ 110$.
- $001 = 1$, $010 = 2$, $110 = 6$. Final answer: 126 (octal).

■ 5 True or False

1. TRUE ($11001 = 25$)
2. FALSE ($7A = 1111010$, not 1110101)
3. TRUE ($57_{10} = 101\ 111$)
4. TRUE
5. TRUE ($1010110 = 126_{10}$)